THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $(x+p)(x+q) = x^2 + 11x + 28$, then $p^2 + q^2$ equals: 1

(a) 65 **(b)** 121 **(c)** 93 **(d)** 177

2. The area of a square of side 36 cm exceeds that of a square of side 32 cm by: 1

(a) 16 **(b)** 144 **(c)** 1024 **(d)** 272

3. The value of $48^2 - 47^2$ is: 1

(a) 97 **(b)** 96 **(c)** 95 **(d)** 48

4. If $a^2 - b^2 = 0$ then: 1

(a) $a = b$ or $a = -b$ **(b)** $a + b = 1$ **(c)** $a = 0$ only **(d)** $a = b$ only

5. If $x + y = 10$ and $xy = 21$, then $x^2 + y^2$ equals: 1

(a) 58 **(b)** 100 **(c)** 79 **(d)** 142

6. If $x - y = 2$ and $xy = 35$, then $x^2 + y^2$ equals: 1

(a) 74 **(b)** 4 **(c)** -66 **(d)** 39

7. The value of $903^2 - 97^2$ is: 1

(a) 806100 **(b)** 824818 **(c)** 806000 **(d)** 649636

8. If $x + \frac{1}{x} = 6$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 35 **(b)** 36 **(c)** 38 **(d)** 34

9. **Assertion (A):** $49 \times 51 = 2499$. 1

Reason (R): $(a+b)(a-b) = a^2 - b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $(3x)^2 = 9x^2$. 1
Reason (R): $(3x)^2 = 3x^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $(x + p)(x + q) = x^2 + 15x + 54$, find $p^2 + q^2$. 2

12. If $x + y = 17$ and $xy = 60$, find $x^2 + y^2$. 2

13. Evaluate $960^2 - 40^2$ using an identity. 2

14. If $(x + p)(x + q) = x^2 + 9x + 14$, find $p^2 + q^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x - y = 7$ and $xy = 30$, find $x^2 + y^2$ and $(x + y)^2$. 3

16. Simplify $(6x + 1)^2 - (6x - 1)^2$. 3

17. If $a + b = 11$ and $ab = 30$, find $a^2 + b^2$ and $(a - b)^2$. 3

OR

If $x - y = 8$ and $xy = 48$, find $x^2 + y^2$ and $(x + y)^2$. 3

18. Simplify $(3x + 8)^2 - (3x - 8)^2$. 3

OR

If $a + b = 14$ and $ab = 45$, find $a^2 + b^2$ and $(a - b)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A stationery shop in Chandigarh sells registers at ₹a each and pens at ₹b each. A bulk pack has 12 registers and 18 pens, so its bill is $12a + 18b$ rupees. Rahul wants to simplify the bill by factorisation. 5
(i) Find the highest common factor of 12 and 18.
(ii) Factorise $12a + 18b$.
(iii) Find the bill when $a = 15$ and $b = 10$.
(iv) Factorise $4x^2 - 36$ completely and verify at $x = 5$.

20. The sides of a rectangle are $(7x + 2)$ m and $(7x - 2)$ m. Find its area as an expression, its value at $x = 4$, and by how much it is less than the area of a square of side $7x$ m. 5

OR

If $x + y = 15$ and $x^2 + y^2 = 173$, find xy, $(x - y)^2$ and $x - y$ (take $x > y$). 5

END OF QUESTION PAPER — Total: 40 Marks